

Solution Stoichiometry

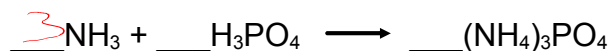
The only difference between solution and gravimetric stoich is how we find moles:

$$n = CV \quad \text{instead of} \quad n = \frac{m}{M}$$

G $n = \frac{m}{M}$ ratio $m = nM$
 mass $\rightarrow$ moles $\rightarrow$ moles $\rightarrow$ mass

V $n = CV$ ratio $V = \frac{n}{C}$
 volume $\rightarrow$ moles $\rightarrow$ moles $\rightarrow$ volume

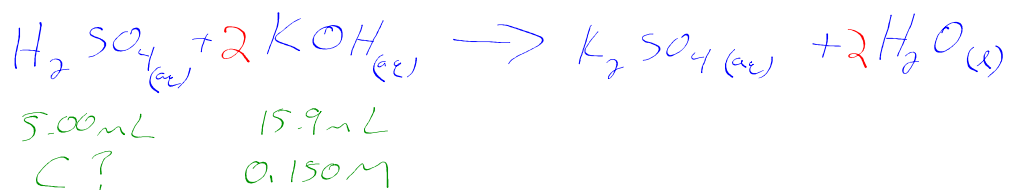
Ex) What volume of 15.0 M $\text{NH}_3(\text{aq})$ would be required to react completely with 1.5 L of 12.9 M $\text{H}_3\text{PO}_4(\text{aq})$ according to the following reaction?



$$V_{\text{NH}_3} = 1.5 \text{ L H}_3\text{PO}_4 \times \frac{12.9 \text{ mol H}_3\text{PO}_4}{1 \text{ L}} \times \frac{3 \text{ mol NH}_3}{1 \text{ mol H}_3\text{PO}_4} \times \frac{1 \text{ L NH}_3}{15.0 \text{ mol}}$$

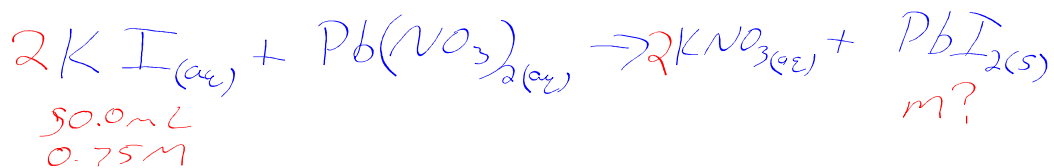
$$= 3.9 \text{ L}$$

Ex) In an experiment, a 5.00 mL sample of sulfuric acid reacts completely with 15.9 mL of 0.150 M potassium hydroxide solution. Calculate the molar concentration of sulfuric acid.



$$\begin{aligned} C_{\text{H}_2\text{SO}_4} &= 0.159 \cancel{\text{L KOH}} \times \frac{0.150 \cancel{\text{mol KOH}}}{1\text{ L}} \times \frac{1\text{ mol H}_2\text{SO}_4}{2\cancel{\text{mol KOH}}} \times \frac{1}{0.00500\text{ L H}_2\text{SO}_4} \\ &= 0.239\text{ mol/L} \end{aligned}$$

Ex) 50.0 mL of 0.75 M potassium iodide is reacted with excess lead (II) nitrate. What mass of precipitate would you expect to recover?



$$\begin{aligned} m_{\text{PbI}_2} &= 0.050\text{ L KI} \times \frac{0.75\text{ mol KI}}{1\text{ L}} \times \frac{1\text{ mol PbI}_2}{2\text{ mol KI}} \times \frac{461.00\text{ g}}{1\text{ mol PbI}_2} \\ &= 8.6\text{ g} \end{aligned}$$

Ex) In a lab experiment, 125 mL of 0.70 M potassium iodide is poured into a beaker containing 90.0 mL of 0.65 M lead (II) chloride. 17.5 g of yellow precipitate is recovered. What is the percent yield of this reaction?



Assume KI is limiting

$$V_{\text{PbCl}_2} = 0.125 \text{ L KI} \times \frac{0.70 \text{ mol KI}}{\text{L}} \times \frac{1 \text{ mol PbCl}_2}{2 \text{ mol KI}} \times \frac{1 \text{ L PbCl}_2}{0.65 \text{ mol}}$$

$$= 0.067 \text{ L}$$

or
67 mL

$\therefore$ KI is limiting

$$m_{\text{PbI}_2} = 0.125 \text{ L KI} \times \frac{0.70 \text{ mol KI}}{\text{L}} \times \frac{1 \text{ mol PbI}_2}{2 \text{ mol KI}} \times \frac{461.00 \text{ g}}{\text{mol}}$$

$$= 2.0 \times 10 \text{ g}$$

$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100$$

$$= \frac{17.5}{20} \times 100$$

$$= 88\%$$

1) 0.430 M

2) 72.0 mL

3) 0.157 M

4) 0.500 ML

1) 1330 mL

2) 1.0×10^2 mL

3) 5.73 g

4) 2.87 g

5) NO, 8.88 g